1. As part of the machine leering process, it is necessary to perform data pro-processing, which involves missing values handling and features encoding, assuming you are provided with the following data.

Customer Name	Gender	Age	Payment Method	Last Transaction	Churn
Jeffrey Pierce	male	46	credit card	86	loyal
Marion Wade	female	71	credit card	27	loyal
Tim Harrison	male	68	cash	179	loyal
Grace Newton			cash	173	churn
Nora Holmes	female	4	cash	188	churn
Tracy Osborne	female	23		118	loyal
Ricardo Jordan	male	38	credit card	131	loyal
Tammy Kennedy		65	cash	187	churn
Ellis Lloyd	female		credit card	72	loyal
Marvin Silva	male	21	credit card	8	loyal
Christine Simon	female	37		162	churn
Franklin Griffith	female	70	credit card	153	churn
Juan Barnett	male	81	cash	153	churn
Velma Montgomery			credit card	210	churn
Mamie Cox	male	55	cash	126	churn
June Patrick	male	42	credit card	83	loyal
Edgar Frank	male	38	credit card	114	loyal
Frances Benson		60		142	loyal
Billie Thomson	male		credit card	112	loyal
Roman Patterson	male	22	credit card	88	loyal

a. Explain the missing value handling method that you use and provide an overview of the data after the missing values have been handled.
Answer :
The appropriate method to handle the missing values is to change the missing values on the data to it's mean for numeric variables and mode for the categorical variables.

Customer Name	Gender	Age	Payment Method	Last Transaction	Churn
Jeffrey Pierce	male	46	credit card	86	loyal
Marion Wade	female	71	credit card	27	loyal
Tim Harrison	male	68	cash	179	loyal
Grace Newton	male	46	cash	173	churn
Nora Holmes	female	4	cash	188	churn
Tracy Osborne	female	23	credit card	118	loyal
Ricardo Jordan	male	38	credit card	131	loyal
Tammy Kennedy	male	65	cash	187	churn
Ellis Lloyd	female	46	credit card	72	loyal
Marvin Silva	male	21	credit card	8	loyal
Christine Simon	female	37	credit card	162	churn
Franklin Griffith	female	70	credit card	153	churn
Juan Barnett	male	81	cash	153	churn

Velma Montgomery	male	46	credit card	210	churn
Mamie Cox	male	55	cash	126	churn
June Patrick	male	42	credit card	83	loyal
Edgar Frank	male	38	credit card	114	loyal
Frances Benson	male	60	credit card	142	loyal
Billie Thomson	male	46	credit card	112	loyal
Roman Patterson	male	22	credit card	88	loyal

b. Explain the feature encoding method for categorical data, and provide an overview of the data after being encoded.
Answer : The Feature encoding method is to change the categorical data to integer counterpart.
   i. male: 0; female: 1
   ii. credit card: 0; cash: 1
   iii. churn: 0; loyal : 1

Customer Name	Gender	Age	Payment Method	Last Transaction	Churn
Jeffrey Pierce	0	46	0	86	1
Marion Wade	1	71	0	27	1
Tim Harrison	0	68	1	179	1
Grace Newton	0	46	1	173	0
Nora Holmes	1	4	1	188	0
Tracy Osborne	1	23	0	118	1
Ricardo Jordan	0	38	0	131	1
Tammy Kennedy	0	65	1	187	0
Ellis Lloyd	1	46	0	72	1
Marvin Silva	0	21	0	8	1
Christine Simon	1	37	0	162	0
Franklin Griffith	1	70	0	153	0
Juan Barnett	0	81	1	153	0
Velma Montgomery	0	46	0	210	0
Mamie Cox	0	55	1	126	0
June Patrick	0	42	0	83	1
Edgar Frank	0	38	0	114	1
Frances Benson	0	60	0	142	1
Billie Thomson	0	46	0	112	1
Roman Patterson	0	22	0	88	1

3. The regression data in the table below consists of 7 data samples that are utilized as training data for the K-Nearest Neighbor (KNN) Algorithm.

ID	Years Experience	Age (year)	Salary (IDR)
1	1.2	30	5901600.00
2	1.4	22	6930900.00
3	1.6	32	5659800.00
4	2.1	25	6528900.00
5	2.3	33	5983800.00
6	3	40	8496450.00
7	3.1	37	9022650.00
8	3.3	29	?

a. Apply the MinMax Scaler technique to normalize the Age and Salary features.

Formula : $y = \frac{x - \min}{\max - \min}$

Age :

$$min = 22; max = 40; (max - min) = 18$$

ID	Age (year)	Scaled Age	Calculation
1	30	0.44	$\frac{30-22}{18}$
2	22	0	$\frac{22-22}{18}$
3	32	0.56	$\frac{32-22}{18}$
4	25	0.17	$\frac{25-22}{18}$
5	33	0.61	$\frac{33-22}{18}$
6	40	1	$\frac{40-22}{18}$
7	37	0.83	$\frac{37-22}{18}$
8	29	0.39	$\frac{29-22}{18}$

Salary :

$$min = 5659800; max = 9022650; (max - min) = 3362850$$

ID	Salary (IDR)	Scaled Salary	Calculation
1	5901600.00	0.07	$\frac{5901600-5659800}{3362850}$
2	6930900.00	0.38	$\frac{6930900-5659800}{3362850}$
3	5659800.00	0	$\frac{5659800-5659800}{3362850}$
4	6528900.00	0.26	$\frac{6528900-5659800}{3362850}$
5	5983800.00	0.1	$\frac{5983800-5659800}{3362850}$

6	8496450.00	0.84	$\frac{8496450 - 5659800}{3362850}$
7	9022650.00	1	$\frac{9022650 - 5659800}{3362850}$

b. K-Regression
With Scaled Data

ID	Years Experience	Age (year)	Salary (IDR)	Distance	Calculation
7	3.1	0.83	1.00	0.483	$\sqrt{(3.3-3.1)^2+(0.39-0.83)^2}$
6	3	1	0.84	0.68	$\sqrt{(3.3-3)^2+(0.39-1)^2}$
5	2.3	0.61	0.10	1.024	$\sqrt{(3.3-2.3)^2+(0.39-0.61)^2}$
4	2.1	0.17	0.26	1.22	$\sqrt{(3.3-2.1)^2+(0.39-0.17)^2}$
3	1.6	0.56	0.00	1.708	$\sqrt{(3.3-1.6)^2+(0.39-0.56)^2}$
2	1.4	0	0.38	1.94	$\sqrt{(3.3-1.4)^2+(0.39-0)^2}$
1	1.2	0.44	0.07	2.101	$\sqrt{(3.3-1.2)^2+(0.39-0.44)^2}$

Regression:

$$y = \frac{\sum_{i=1}^{n=3} y_n}{n} = \frac{1+0.84+0.1}{3} = \frac{1.94}{3} = 0.65$$

DeScaling:

$$0.65 = \frac{x - 5659800}{3362850}$$
$$x = 7845652$$

So the Value for the 8th data point is 7845652

With Original Data (KALO MAU NYONTEK AMBIL YG INI AJA BUAT YG 3b. TAKUTNYA YG ATAS SALAH)

ID	Years Experience	Age (year)	Salary (IDR)	Distance	Calculation
1	1.2	30	5901600.00	2.326	$\sqrt{(3.3-3.1)^2+(29-30)^2}$
3	1.6	32	5659800.00	3.448	$\sqrt{(3.3-3)^2+(29-32)^2}$
5	2.3	33	5983800.00	4.123	$\sqrt{(3.3-2.3)^2+(29-33)^2}$
4	2.1	25	6528900.00	4.176	$\sqrt{(3.3-2.1)^2+(29-25)^2}$
2	1.4	22	6930900.00	7.253	$\sqrt{(3.3-1.6)^2+(29-22)^2}$
7	3.1	37	9022650.00	8.002	$\sqrt{(3.3-1.4)^2+(29-37)^2}$
6	3	40	8496450.00	11.004	$\sqrt{(3.3-1.2)^2+(29-40)^2}$

Regression:

$$y = \frac{\sum_{i=1}^{n=3} y_n}{n} = \frac{5901600+5659800+5983800}{3} = \frac{17545200}{3} = 5848400$$

So the Value for the 8th data point is 5848400

4. Confusion Matrix

		Predicted			
		A	B	C	D
Actual	A	7	2	1	0
	B	0	8	1	0
	C	1	0	5	0
	D	0	0	1	4

Precision:

$$Precision = \frac{TP}{FP + TP}$$

A. $A_{prec} = \frac{7}{7+1} = \frac{7}{8} = 0.875$
B. $B_{prec} = \frac{8}{2+8} = \frac{8}{10} = 0.8$
C. $C_{prec} = \frac{5}{5+3} = \frac{5}{8} = 0.625$
D. $D_{prec} = \frac{4}{4} = 1$

Sensitivity

$$Sensitivity = \frac{TP}{FN + TP}$$

A. $A_{sens} = \frac{7}{7+3} = \frac{7}{10} = 0.7$
B. $B_{sens} = \frac{8}{8+1} = \frac{8}{9} = 0.89$
C. $C_{sens} = \frac{5}{5+1} = \frac{5}{6} = 0.83$
D. $D_{sens} = \frac{4}{4+1} = \frac{4}{5} = 0.8$

F1 Score

$$F1\ Score = 2 * \frac{Precision * Sensitivity}{Precision + Sensitivity}$$

A. $A_{F1} = 2 * \frac{0.875*0.7}{0.875+0.7} = 2 * \frac{0.6125}{1.575} = 2 * 0.39 = 0.78$
B. $B_{F1} = 2 * \frac{0.8*0.89}{0.8+0.8} = 2 * \frac{0.712}{1.69} = 2 * 0.42 = 0.84$
C. $C_{F1} = 2 * \frac{0.625*0.83}{0.625+0.83} = 2 * \frac{0.5188}{1.455} = 2 * 0.36 = 0.72$
D. $D_{F1} = 2 * \frac{1*0.8}{1+0.8} = 2 * \frac{0.8}{1.8} = 2 * 0.44 = 0.88$

Accuracy

$$Accuracy = \frac{TP}{Count\ Occurance} = \frac{\sum_{i=1}^{n} F1Score_i}{n}$$

$$Accuracy = \frac{0.78 + 0.84 + 0.72 + 0.88}{4} = 0.8$$